



The Crisis

50%

NCDs drive over half of deaths in India—hypertension, diabetes, COPD dominate

9/10

Indians have inadequate health literacy —patients forget medications, diet, follow-ups follow-ups



PHCs become "repetition factories" —same uncontrolled cases cycle through with poor self-care



The Opportunity

1.5L

Ayushman Bharat HWCs scaling to 1.5 lakh sites as chronic care backbone



Digital health digitizes records but rarely transforms them into patient-friendly instructions



JanMitra bridges the last-mile gap —converting PHC visits into actionable care plans



How It Works

1

Capture

Staff snaps prescription photo or records 15-30s voice note in local language

2

Extract & Structure

AI extracts medicines, doses, timings, advice, follow-up dates automatically

3

Generate Care Plan

Plain-language schedule: "Morning—take X mg before food; Evening—walk 30 min"

4

Deliver Instantly

WhatsApp, SMS, printed slip, or voice call—adapted to patient's access



Deployment Contexts

Ayushman Arogya Mandirs

NCD Clinics

ASHA Home Visits

Proposed Method & System Architecture

GenAI-powered multimodal processing with medical RAG and safety guardrails

Multimodal Inputs

Prescription images (handwritten/printed)

Voice notes from doctors/ANMs

Optional text from EMR systems

AI Processing Stack

OCR + Structured Extraction

Robust OCR for Indian scripts + LLM-assisted parsing of drugs, doses, advice

Medical RAG Engine

Retrieval over ICMR/WHO/AIIMS guidelines—grounded explanations, not hallucinations

Multilingual LLM Generator

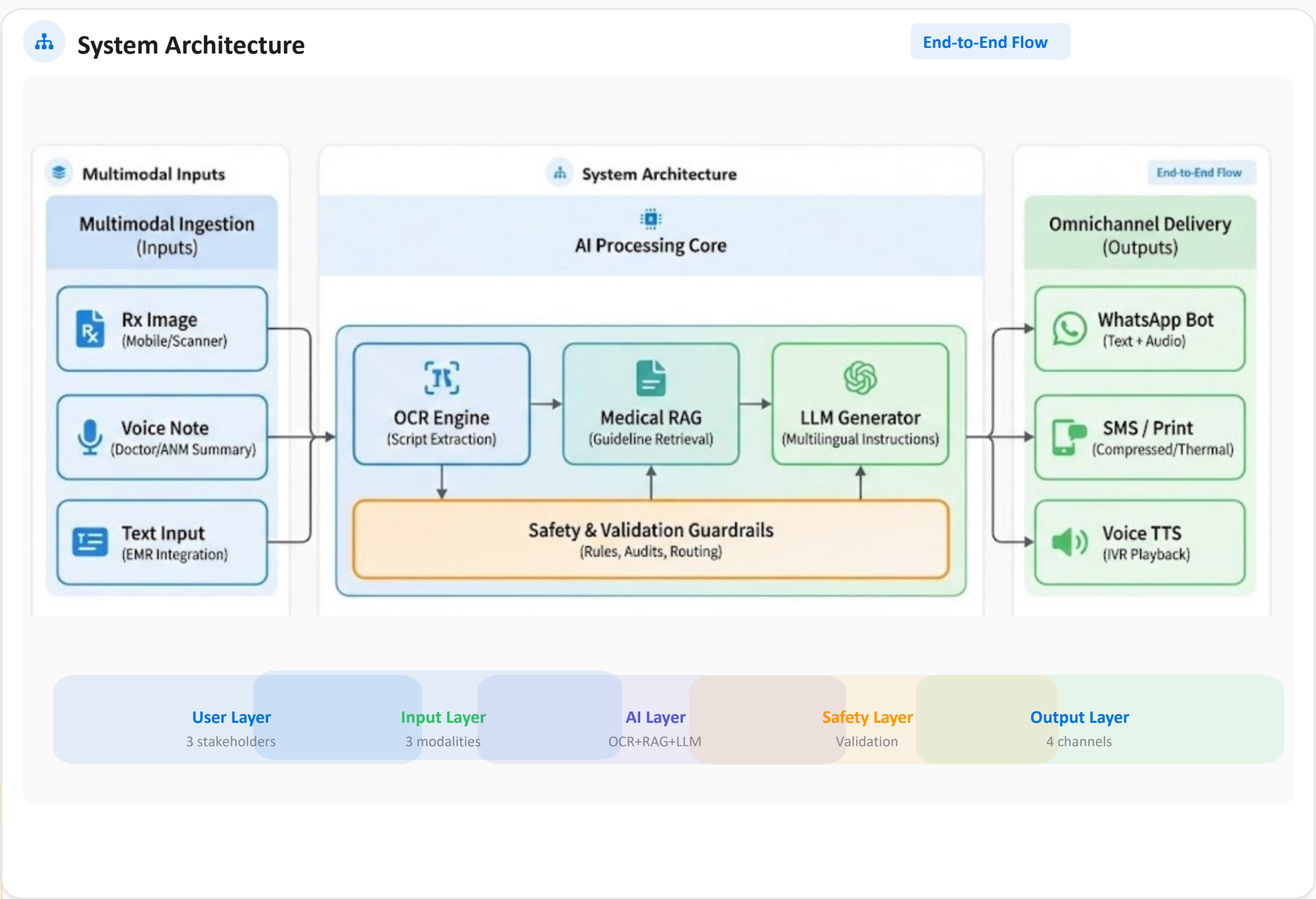
Parallel outputs in English + local language with culturally appropriate framing

Safety Guardrails

Medicines/doses never altered—only rephrased & scheduled

No new drugs, no diagnosis beyond prescription

Low-confidence cases flagged for human review



Datasets, Experiments, Novelty & Scale

Evidence-based development with rigorous evaluation and national scalability

Data Sources

Clinical Guidelines

ICMR, MoHFW, AIIMS, WHO protocols for NCDs as RAG corpus

Prescription Data

Synthetic & de-identified PHC/NCD clinic samples for training

Language Resources

Regional phrase banks (Hindi, Tamil) + doctor→patient parallel corpora

Evaluation

1 Doctor Validation

PHC MOs review for correctness, omission, safety

2 Patient Comprehension

Pre-post quiz + teach-back method

3 Adherence Tracking

Self-report checklists, IVR, refill patterns

Hallucination Audit

Quantitative audit driving rate to near-zero

Novelty

Beyond Generic Chatbots

Anchored in actual PHC workflows and prescriptions—not open Q&A. Clear guardrails: structures & translates only, never diagnoses or prescribes.

India-First Design

Tailored for Ayushman Bharat HWCs as first point of chronic care. Built for low-literacy, multi-lingual, low-resource settings including non-smphone households.

Ecosystem Integration

Ready for ABDM health IDs, AB-HWC apps, state HIS platforms via APIs. Language & guideline "plug-ins" enable rapid state-wise rollout.

Target Impact Metrics

30%

↓ Repeat visits

2x

↑ Comprehension

50%

↑ Follow-up

<1%

Hallucination

Scalability

Language Extensibility

Swap language packs for pan-India rollout

Guideline Plugins

State-specific protocols with minimal code change

API-First Design

ABDM, AB-HWC, state HIS integration ready

Multi-Channel Output

WhatsApp, SMS, print, voice—universal access

National Vision

- ✓ Nationwide deployment across 1.5L+ Ayushman Arogya Mandirs
- ✓ Anonymized analytics on adherence, follow-up patterns, disease burden
- ✓ PHC-level insights for public health planning & resource allocation
- ✓ Transforming PHCs from repetition factories to effective care hubs